

COURSES OF RELEVANCY FOR ETH ZURICH

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FALL 2021

MATH 250: Linear Algebra. Introduction to matrices, vector spaces, linear transformations, determinants, eigenvalues, and eigenvectors, and their applications to systems of linear equations. Used Leon, *Linear Algebra with Applications*.

PHYS 151: Seminar Modern Physics. First course in modern physics covering special relativity, early quantum theory including one-dimensional potentials and the Schrödinger equation, and basic thermodynamics. Included phenomena like photoelectric effect, Bohr and Rutherford atomic models, and quantum tunneling. Had a lab section. Used Townsend, *A Modern Approach to Quantum Mechanics*.

SPRING 2022

ASTR 211: Observing and Data Analysis. Astronomy course focused on observational techniques, telescope use, and quantitative data analysis. We studied Fourier transforms and their use in radio astronomy. This course included a lab section where we used our university telescope.

MATH 350: Real Analysis. Rigorous treatment of single-variable analysis covering open and closed sets, limits, continuity, compactness, differentiation, Riemann integration, monotone convergence, sequences and series, and convergence of function sequences. We also covered construction of the real numbers and discussed the Peano axioms. We used Silva, *An Introduction to Real Analysis*

FALL 2023

On Medical Leave. I was on leave for this semester due to a ski accident, where I dislocated both shoulders, and received surgery.

SPRING 2023

MATH 413: Algebraic Geometry. Upper-level mathematics course introducing affine and projective varieties, Noetherian rings, and developed the correspondence between polynomial ideals and geometric objects. We proved Hilbert's Nullstellensatz. This course was physics-based.

MATH 453: Partial Differential Equations. First course on classical linear PDEs in two and three dimensions such as the heat, wave, and Laplace equations, covering derivations, boundary-value problems, separation of variables, and Fourier methods. This course was physics-based.

PHYS 202: Vibrations, Waves and Optics. Intermediate physics course treating Fourier series and Fourier transforms in one and two dimensions, oscillations, in circuits and mass-springs systems, wave propagation through varying media, interference and diffraction, geometric optics, and topics in advanced optics. This course included a lab component. Based on Smith, *Waves and Oscillations*.

SUMMER 2023

MATH 300: Advanced Abstract Algebra. At BSM. An upper-division algebra course covering group actions, Burnside's lemma, the three Sylow theorems, properties of nilpotent and solvable groups, finite fields and Frobenius automorphisms, and modules over rings.

MATH 310: Real Functions and Measures. At BSM. A measure theory course introducing σ -algebras, Lebesgue and Borel measure spaces, Lebesgue integration, Dominated Convergence, function spaces such as L^p , and Fourier transform.

MATH 325: Complex Functions. At BSM. An introductory complex analysis course covering open sets, continuity, holomorphic functions, the Cauchy–Riemann equations, contour integration, Rouché's theorem, and the Schwarz lemma, including discussion of consequences of the Jordan curve theorem. We used Needham, *Visual Complex Analysis*

MATH 449: Research Opportunities. At BSM. Non-credit research-oriented seminar studying 3-partite hypergraphs. My name is acknowledged in the paper that resulted from this class.

FALL 2023

MATH 383: Complex Analysis. Introductory course on functions of a complex variable, covering holomorphic functions, Cauchy's integral theorem, power series, residues, and applications such as conformal mappings. The course concluded with special topics including proofs of the Prime Number Theorem and the Central Limit Theorem, as well as a unit on Fourier and Laplace transforms, following Stein and Shakarchi's *Complex Analysis*.

MATH 407: Dance of the Primes. Advanced number theory course on prime numbers and their distribution, exploring topics like the prime number theorem, zeta functions, and related analytic methods.

PHYS 201: Electricity and Magnetism. Core course covering electrostatics, magnetostatics, electromagnetic fields, Maxwell's equations, circuits, power, and electromagnetic waves. This lab had a lab component covering AC and DC circuits through LC, LRC, and circuits involving amplifiers and transistors. This course followed Purcell, *Electricity and Magnetism*.

PHYS 301: Quantum Physics. Upper-level introduction to nonrelativistic quantum mechanics. Beginning with a discussion of the Schrödinger wave equation, the concept of matter waves, and wave-packets, one-dimensional problems such as the particle-in-a-box and the harmonic oscillator, and discussion of the structure of the hydrogen atom. The course ends with angular momentum and spin, with applications to atomic physics, entanglement, and quantum information. This course has a lab section, where experiments detecting single photons, observing hyperfine spectra, etc.

SPRING 2024

MATH 126: Introduction to Partial Differential Equations. At University of California Berkeley. Course on classical PDEs in n -dimensions. Covering the heat equation, wave equation, transport equation, integrating by parts in higher dimensions, and the use of energy to solve PDEs. Course used rigorous analysis to show the Maximum Principle for the heat equation, and energy of the wave equation. Based on Evans, *PDEs*

MATH 141: Elementary Differential Topology. At the University of California, Berkeley, completed a course on smooth manifolds covering tangent spaces, smooth maps, basic transversality (including Thom's transversality theorem) and embedding theorems. The course introduced intersection numbers, orientations, and tensors, then developed oriented intersection theory and included proofs of the Hopf degree theorem and the Gauss–Bonnet theorem, following Guillemin and Pollack's *Differential Topology*.

MATH 205: Theory of Functions of a Complex Variable. Graduate-level second-semester complex analysis course at the University of California, Berkeley, focused on conformal mappings, modular forms, Haar measure, complex manifolds, and additional advanced topics. The course followed Ahlfors' *Complex Analysis* as the primary text.

FALL 2024

MATH 493: Senior Thesis in Mathematics. Independent mathematics research project done with Professor Ivo Terek culminating in a written thesis and oral presentation on a Delzant-type correspondence for Hamiltonian $SO(3)$ actions. The class was based on symplectic topology from Ana Cannas da Silva's *Lectures in Symplectic Geometry* and differential geometry using John Lee's *Differentiable Manifolds*, and included solving problems with Professor Ivo Terek on differential geometry, Lie theory, and symplectic geometry. Topics learned included symplectic forms, the exponential map, Lie groups and Lie algebras, group actions on manifolds, adjoint and coadjoint actions, Hamiltonian functions on manifolds, and Hamiltonian vector fields.

MATH 497: Topics in Measure and Ergodic Theory. Advanced graduate topics course with Professor Cesar Silva on measure theory and dynamical systems, covering measure spaces, measure-preserving transformations, measure theoretic pullbacks, Radon–Nikodym Theorem, product constructions for weak mixing, and related notions of algebras and the Monotone Class theorem. The course developed the entropy of a transformation T , and included an independent project in which I studied and developed the theory of Lyapunov exponents.

PHYS 411T: Classical Mechanics. Advanced mechanics course using Lagrangian and Hamiltonian formalisms to study single- and many-particle systems, rigid bodies, small oscillations, chaotic oscillations, continuum mechanics and tensors and their applications. This course also introduced the theory of Lagrange multipliers for physical constraints. Based on Taylor, *Classical Mechanics*.

SPRING 2025

MATH 494: Senior Thesis in Mathematics. Final semester of the mathematics thesis project. Done under supervision of Professor Ivo Terek. Classified images of moment map μ_c arising from the Lie-group action $\mathrm{SO}(3) \curvearrowright \mathcal{M}$ as either a ball or slicings of a ball, and presented the results in the thesis defence.

MATH 497: Geometry and Quantum Theory. Independent graduate-level reading course in mathematical physics with Professor Ivo Terek, covering Lagrangian mechanics, symplectic geometry, axiomatic quantum field theory, Lie group actions and semidirect products, fiber bundles, and geometric quantization. The course was based on Daniel Freed’s *Quantum Theory from a Geometric Viewpoint*.

PHYS 302: Statistical Mechanics and Thermodynamics. Upper-level course in thermodynamics and statistical mechanics, developing concepts like temperature, entropy, thermodynamic functions such as Gibbs free energy. Topics included the partition function, grand canonical ensemble, Ising model, Fermi energy, semiconductors, and the theory of phase transitions and critical temperatures. Based on Schroeder, *An Introduction to Thermal Physics*.

PHYS 493: Senior Thesis in Physics. During the first semester of my physics thesis, I analyzed 180m Ta data from the MAJORANA DEMONSTRATOR experiment at the Sanford Underground Research Facility in South Dakota using likelihood-based techniques. I learned frequentist statistical methods, Monte Carlo simulations, and the maximum likelihood principle. I also learned to use the Geant4 toolkit to model particle interactions.

MATH 433: Mathematical Modeling. Advanced course on constructing and analyzing mathematical models for real-world phenomena. This course covered Euler's Method, RK4, the method of finite differences, Markov Chains, linear and nonlinear systems, nondimensionalization, with applications to biology, physics, and social sciences. For the final project, I simulated the 2D Wave and 2D Biharmonic equations with a finite element method using tetrahedra.

p -Adic Ergodic Theory. Faculty-supervised independent study focused on p -adic Ergodic theory. One research question is whether there exists a p -adic transformation which is minimal but not uniquely ergodic.

PHYS 418T: Gravity. Upper-level physics course on gravity and general relativity. Topics include the geometrical formulation of special relativity; the equivalence principle and motivation for gravity as spacetime curvature; tensors in curved spacetime and the Einstein field equations; the Schwarzschild geometry and solar-system tests of GR; and applications of GR to black holes, gravitational waves, and the expanding universe.

PHYS 494 – Senior Research: Physics. During the first semester of my physics thesis, I finished the analysis of 180m Ta data from the MAJORANA DEMONSTRATOR experiment. I wrote my findings in my senior thesis, and will give a colloquium in January.